Day 1

NLP & Text Preprocessing

Core concepts explained visually

Raw Text → Tokenize → Clean → Normalize → Clean Text

The Full NLP Pipeline

Where today's concepts fit in the bigger picture

1. Text Preprocessing

TODAY

Tokenization

"great!" → [great, !]

Lowercasing

GOOD → good

Punctuation Removal

great!!! → great

Stop-word Removal

the movie is good → movie good

Lemmatization

running → run

2. Text Representation

LATER

TF-IDF

score words by how important
they are across documents

Embeddings

represent each word as a
list of numbers (a vector)

3. Model

LATER

Classification / etc.

use the representation to
predict a label, e.g. sentiment

Today, we build the foundation: Text Preprocessing. TF-IDF, Embeddings, and Modeling come later.

The Big Picture

What Day 1 is really about

One idea to remember

Human text is meaningful to us — but messy and inconsistent for a machine.

RAW TEXT
"The MOVIE is NOT good!!!"

→

TOKENIZE
[The, MOVIE, is, NOT, good, !!!]

→

CLEAN
[the, movie, is, not, good]

→

NORMALIZE
[movie, not, good]

→

READY TEXT
"movie not good"

Goal: remove unnecessary variation — without removing the meaning the task

1. NLP & Raw Text

Start by understanding what enters the pipeline

Human view

“This movie was absolutely amazing!”

We immediately understand the meaning.

Meaning

Machine view

CAPS	MOVIE
Punctuation	!!!
Numbers	10/10
HTML	
Contractions	wasn't

Raw text = text exactly as it arrives, before preprocessing.

2. Tokenization

Split text into smaller processable units

What is a token?

A token is a unit of text. It can be a word, a sub-word, or sometimes a character.

Word tokenization

The movie was great!

The	movie	was	great	!

Each full word becomes a token.

Sub-word tokenization

playing → play | ing

Useful when a full word is unknown but its smaller pieces are known.

Tokenization SPLITS text — it does not clean it.

3. Cleaning & Normalization

Reduce unnecessary variation in the text

LOWERCASING

Good • GOOD • good

↓

good

Reduces case-based variation.

PUNCTUATION

great! • great!!!

↓

great

Removes punctuation when it is not useful.

NUMBERS

This movie is 10/10

↓

Keep or remove?

Numbers can carry task-specific signal.

Cleaning is a decision: remove noise, preserve useful information.

4. Stop-word Removal

Common words can be low-signal — but not always

Typical stop words

the | is | a | this | was

They may add little signal in some tasks.

Goal: reduce low-value words.

The negation trap

this movie is not good

movie good

Meaning changed!

Task-aware rule: keep “not” when sentiment depends on it.

5. Lemmatization vs. Stemming

Both reduce word variation — but in different ways

running • ran • studies

STEMMING

Mechanical chopping

studies → studi
connected → connect

Fast, but may produce non-words.

LEMMATIZATION

Dictionary-like base form

running → run
ran → run

More meaning-aware; usually real words.

Day 1 final pipeline uses lemmatization.

6. Noise vs. Signal

The most important decision in preprocessing

Does this information help the task?

YES → SIGNAL

KEEP IT

Examples in sentiment:
not • 10/10 • love • terrible

NO → NOISE

REMOVE / NORMALIZE

Examples here:
HTML tags • redundant case • some punctuation

No cleaning rule is correct for every NLP task.

7. The Complete Day 1 Pipeline

See every concept working together

RAW
The MOVIE is NOT good!!!

→

HTML CLEANUP
The MOVIE is NOT good!!!

→

LOWERCASE
the movie is not good!!!

→

TOKENIZE
[the, movie, is, not, good, !!!]

STOP WORDS
[movie, not, good]

→

LEMMA
[movie, not, good]

→

FINAL TEXT
movie not good

Every step has a different job — but the same goal: useful, consistent text.

Day 1 Mental Map

A one-slide revision of the core concepts

1

RAW TEXT

Text before any cleaning.

2

TOKENIZATION

Split text into tokens.

3

CLEANING

Remove selected noise.

4

STOP WORDS

Remove low-signal words carefully.

5

LEMMATIZATION

Reduce words to useful base forms.

THE GOLDEN RULE

Preprocessing is task-dependent: remove noise, preserve signal.

If a cleaning step changes the meaning your task needs, it is the wrong cleaning step.